

8.3.1 Backend Booking Logic (Node.js & Express)

The core backbone of the SHIFTY application is the Booking Service. We have successfully implemented the API endpoint that receives the user's inventory and selected package, validates the request, and stores it in the MongoDB database.

Implementation Logic:

The code below represents the createBooking controller. It first validates that the user exists, then calculates the final estimated price by calling the internal ML service, and finally saves the structured booking object.

Code Snippet 1: Booking Controller (bookingController.js)

```
const Booking = require('../models/Booking');
const User = require('../models/User');
const axios = require('axios'); // Used to communicate with ML Microservice

exports.createBooking = async (req, res) => {
  try {
    const { userId, source, destination, inventory, distance_km } = req.body;

    // 1. Validate User
    const user = await User.findById(userId);
    if (!user) return res.status(404).json({ message: "User not found" });

    // 2. Call ML Service for Dynamic Pricing
    // This connects the Node.js backend to the Python ML Container
    const mlResponse = await axios.post('http://ml-service:5000/predict_price', {
      total_items: inventory.length,
      distance: distance_km,
      fragile_count: inventory.filter(i => i.is_fragile).length,
      move_hour: new Date().getHours()
    });

    const predictedPrice = mlResponse.data.price;
    const recommendedPackage = mlResponse.data.package;

    // 3. Create Booking Record
    const newBooking = new Booking({
      user: userId,
      source_address: source,
      destination_address: destination,
      inventory_list: inventory,
      final_price: predictedPrice,
      selected_package: recommendedPackage,
      status: 'CONFIRMED'
    });

    await newBooking.save();

    res.status(201).json({
      message: "Booking confirmed successfully",
      bookingId: newBooking._id,
      price: predictedPrice,
      package: recommendedPackage
    });

  } catch (error) {
    console.error("Booking Error:", error);
    res.status(500).json({ message: "Internal Server Error" });
  }
};
```